WORKED SOLUTION

STEP III 1996, Question 9

Mechanics • Circular Motion

STEP III 1996, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s3-q9

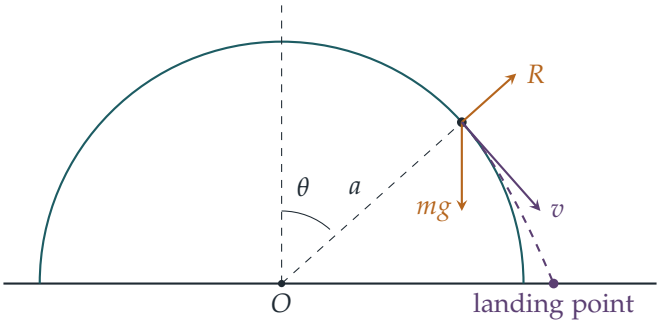
The key idea

The motion has two stages. While the particle stays on the sphere it moves in a circle, so energy conservation gives its speed and the radial equation of motion gives the normal reaction; it leaves the sphere at the moment that reaction drops to zero. After that it is an ordinary projectile, launched along the tangent from a known point with a known speed, and we just need to find where it crosses the level of the centre.

Solution

Model

Treat the particle as a point mass of mass m on a smooth fixed sphere with centre O, with no air resistance. The only forces are its weight mg and the normal reaction R, which acts along the radius away from O. Let θ be the angle between OP and the upward vertical, where P is the particle, and let v be its speed. Since the displacement is small, we take the initial speed to be zero (in the limit).



Stage 1: on the sphere

The reaction is perpendicular to the motion, so it does no work, and energy is conserved. The particle has fallen $a - a\cos\theta$ from the top, so

$$\tfrac{1}{2}mv^2 = mga(1-\cos\theta), \qquad \text{that is} \qquad v^2 = 2ga(1-\cos\theta).$$

While it stays on the sphere it moves in a circle of radius a, so resolving towards O gives

$$mg\cos\theta - R = \frac{mv^2}{a} \quad \Longrightarrow \quad R = mg\cos\theta - 2mg(1-\cos\theta) = mg(3\cos\theta - 2).$$

? **Ask yourself** Why does the particle leave the sphere exactly when $R = 0$?

The sphere can only push, so $R \geq 0$ while there is contact. As θ increases from 0, R decreases steadily from mg, and the formula above would give $R < 0$ beyond $\cos\theta = \frac{2}{3}$. That would need the sphere to pull the particle inwards, which it can't do. So contact is lost when

$$\cos\theta = \tfrac{2}{3}, \qquad \sin\theta = \tfrac{\sqrt{5}}{3}, \qquad v^2 = 2ga(1-\tfrac{2}{3}) = \tfrac{2}{3}ga.$$

Stage 2: the projectile

Take axes with origin O, x horizontal (in the direction of motion) and y vertically up. At the moment of leaving, the particle is at

$$x_0 = a\sin\theta = \tfrac{\sqrt{5}}{3}a, \qquad y_0 = a\cos\theta = \tfrac{2}{3}a,$$

and its velocity is along the tangent, at angle θ below the horizontal:

$$u_x = v\cos\theta = \tfrac{2}{3}v, \qquad u_y = -v\sin\theta = -\tfrac{\sqrt{5}}{3}v, \qquad \text{where } v = \sqrt{\tfrac{2}{3}ga}.$$

Now $y = y_0 + u_y t - \frac{1}{2}gt^2$, so the particle reaches $y = 0$ when

$$\tfrac{1}{2}gt^2 + \tfrac{\sqrt{5}}{3}v\,t - \tfrac{2}{3}a = 0.$$

Taking the positive root,

$$t = \frac{-\frac{\sqrt{5}}{3}v + \sqrt{\frac{5}{9}v^2 + \frac{4}{3}ga}}{g}.$$

With $v^2 = \frac{2}{3}ga$, the expression under the root is $\frac{10}{27}ga + \frac{36}{27}ga = \frac{46}{27}ga$, and $\frac{\sqrt{5}}{3}v = \sqrt{\frac{10}{27}ga}$. Hence

$$t = \sqrt{\frac{a}{27g}}\,(\sqrt{46} - \sqrt{10}).$$

The horizontal distance covered in the flight is

$$\begin{aligned} u_x t &= \tfrac{2}{3}\sqrt{\tfrac{2}{3}ga}\cdot\sqrt{\frac{a}{27g}}\,(\sqrt{46}-\sqrt{10}) = \frac{2\sqrt{2}\,a}{27}(\sqrt{46}-\sqrt{10}) \\ &= \frac{2a}{27}(\sqrt{92}-\sqrt{20}) = \frac{2a}{27}(2\sqrt{23}-2\sqrt{5}) = \frac{a}{27}(4\sqrt{23}-4\sqrt{5}). \end{aligned}$$

Adding the starting distance $x_0 = \frac{\sqrt{5}}{3}a = \frac{9\sqrt{5}}{27}a$, the particle meets the horizontal plane through O at distance

$$x_0 + u_x t = \frac{a}{27}(9\sqrt{5} + 4\sqrt{23} - 4\sqrt{5}) = \frac{a(5\sqrt{5} + 4\sqrt{23})}{27}$$

from the centre, as required.

The flight doesn't touch the sphere

To be thorough, we should check the projectile really is clear of the sphere. Let $f(t) = x^2 + y^2 - a^2$ along the flight path. Expanding in powers of t:

- the constant term is $x_0^2 + y_0^2 - a^2 = 0$;
- the coefficient of t is $2(x_0 u_x + y_0 u_y) = 0$, because the velocity is perpendicular to the radius;
- the coefficient of t^2 is $u_x^2 + u_y^2 - g y_0 = v^2 - ga\cos\theta = 0$, which is exactly the condition $R = 0$;
- what's left is $-g u_y t^3 + \frac{1}{4}g^2 t^4$, and $u_y < 0$.

So $f(t) = g|u_y|\, t^3 + \frac{1}{4}g^2 t^4 > 0$ for $t > 0$. The particle stays outside the sphere, and the landing distance above is the one it actually reaches.

Checks

- The answer is about $1.125a$. This is more than a, as it must be, because the particle has to clear the sphere, which reaches out to distance a at the level of O.
- The leaving height $\frac{2}{3}a$ above the centre is a well-known result, and it doesn't depend on g or m. The landing distance doesn't depend on them either, which dimensional analysis requires: a is the only length in the problem.
- We stepped the motion numerically: a Runge–Kutta integration of $\ddot{\theta} = (g/a)\sin\theta$ from a tiny displacement, stopped when $R = 0$ and continued as a projectile, lands at $1.12458a$. The exact formula gives $1.124580\ldots a$.

Common slips

- Using $v^2 = ga\cos\theta$ (from $R = 0$) but then forgetting to combine it with energy, or using the wrong speed: at leaving, $v^2 = \frac{2}{3}ga$, not $\frac{1}{3}ga$.
- Launching the projectile horizontally. The velocity is along the tangent, at angle θ *below* the horizontal, so it has a downward component from the start.
- Measuring the landing distance from the point of leaving rather than from the centre, which forgets x_0.

Beyond STEP

The condition "contact ends when the reaction vanishes" is an example of a *unilateral constraint*. These appear throughout mechanics, from a bead on a wire to a car going over a hump-backed bridge. The coincidence in the last step of the solution, where the projectile's path and the sphere agree to second order at the point of leaving, says that at that moment the parabola and the circle have the same curvature. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats circular motion, constraint forces and energy methods in this spirit.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.